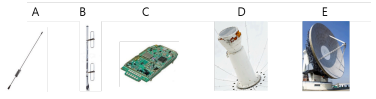
← Back Graded Quiz
Graded Assignment • 30 min

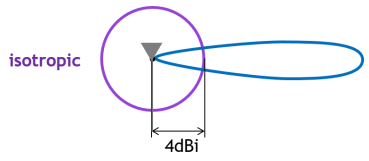
Due Mar 20, 11:59 PM CET

1. Which correctly names the types of antennas shown below? 1 point



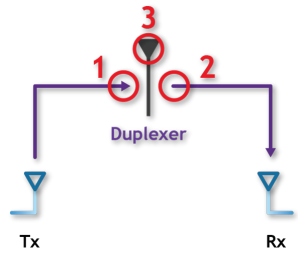
- A: Whip B: PCB C: Feedhorn D: Monopole E: Dipole
- A: Chip B: Dipole C: Microstrip D: Cassegrain E: Monopole
- A: Monopole B: Dipole C: PCB D: Feedhorn E: Cassegrain
- A: Whip B: Monopole C: PCB D: Dipole E: Cassegrain

2. What cannot be the antenna gain for the directed antenna below in dBi? 1 point



- 5 dBi
- 3 dBi
- 8 dBi
- 10 dBi

3. Given the antenna configuration shown below, if we want to send the signal from the Tx to the Rx, which of the following should have high value for good performance? (S is the S-parameter) (Choose two) 1 point



- S11
- S31
- S32
- S21

4. If antenna gain of 10 dBi was achieved using a single antenna, what would be the ideal gain of an array of antennas under identical conditions if the signal power was boosted by 100 times? 1 point

- 50 dBi
- 1000 dBi
- 20 dBi
- 30 dBi
- 15 dBi

5. Which of the statements is erroneous? (Choose two) 1 point

- The directivity of an antenna is determined by the beamwidth.
- Circular array and linear array generate identical radiation patterns.
- Radiation pattern can be only controlled by amplitude distribution.
- With very closely spaced receiver antennas (perfectly correlated channels) for SIMO (single-input-multiple-output), an array gain of 3dB can be achieved.
- Array gain can be used to reduce signal interference to other users.

6. For a perfectly uncorrelated (multipath) channel with 2 Tx antennas and 2 Rx antennas, how many paths can we exploit at maximum? 1 point

- 2
- 1
- 8
- 4
- 6

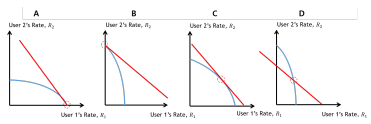
7. If we have a 2 by 2 MIMO system with modulation scheme of 2 bps/Hz per symbol, what is the achievable maximum spectral efficiency? 1 point

- 8 bps/Hz
- 2 bps/Hz
- 1 bps/Hz
- 4 bps/Hz

8. Which of the statements is erroneous? (Choose two) 1 point

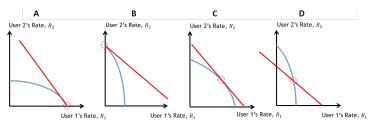
- Diversity can be achieved in space, time, and frequency.
- Tx diversity can be the same as Rx diversity.
- Multi-user MIMO can only be formed when a base station and a user both have multiple antennas at their sides.
- Diversity gain can be made by antenna selection.
- Alamouti coding is a type of antenna selection.

9. Which of the following does not achieve maximum sum rate? 1 point



- A
- B
- C
- D

10. Which of the following is a non-degraded channel? 1 point



- A
- B
- C
- D